

NOVASHYLD TECHNOLOGIES

Cybersecurity Analyst Internship Program

TASK — CYBERSECURITY LAB SETUP

Building a Virtual Cybersecurity Lab Environment for Hands-On Security Training

Prepared By:

Vikram Prajapat

Cybersecurity Analyst Intern

Submission Date: 22-07-2026

NovaShyld Technologies

Internal Training & Development Division

Table of Contents

1. Objective
 2. About the Task
 3. Why It Matters
 4. Lab Requirements
 5. Software Used.
 6. Installation Process
 7. Network Configuration
 8. Lab Verification
 9. Findings
 10. Screenshot Placeholders
 11. Recommendations
 12. Deliverables Checklist
 13. Skills Gained
 14. GitHub Repository Structure
 15. References
-

1. Objective

The objective of this task is to design, build, and validate a fully functional, isolated cybersecurity lab environment that can be used for safe and legal hands-on security practice. As part of the NovaShyld Technologies Cybersecurity Analyst Internship Program, interns are required to develop practical, real-world skills that go beyond theoretical knowledge. A personal cybersecurity lab provides a controlled, sandboxed environment in which an intern can practice offensive and defensive security techniques — including vulnerability scanning, penetration testing, network traffic analysis, and web application security testing — without any risk to production systems, third-party infrastructure, or personal devices.

This report documents the complete process of setting up a virtualized cybersecurity lab, including hardware and software prerequisites, installation procedures, network configuration, verification steps, and the practical findings and skills gained from completing the exercise. The end goal is a repeatable, well-documented lab build that can be referenced by future interns and used as a foundation for subsequent, more advanced internship tasks such as vulnerability assessment, exploitation, and incident response simulations.

Specifically, this task aims to achieve the following outcomes:

- Install and configure a Type-2 hypervisor (VirtualBox) capable of hosting multiple virtual machines simultaneously.
- Deploy an attacker workstation (Kali Linux) pre-loaded with industry-standard penetration testing tools.
- Deploy one or more deliberately vulnerable target machines (Metasploitable2, DVWA, OWASP Broken Web Apps) for safe practice.
- Deploy a Windows 10/11 virtual machine to simulate a realistic corporate endpoint.
- Configure an isolated virtual network so that all lab traffic remains contained and cannot affect the host machine or external networks.
- Verify lab connectivity and functionality using standard networking commands and tools.
- Document findings, screenshots, and lessons learned in a professional report suitable for internal training records.

2. About the Task

2.1 Why a Cybersecurity Lab Is Required

Cybersecurity is fundamentally a practical discipline. While reading about attack techniques, vulnerabilities, and defensive controls builds conceptual understanding, true competency only develops through repeated, hands-on practice. However, practicing security testing techniques — such as port scanning, exploitation, or traffic interception — directly against production systems, third-party websites, or public infrastructure is both illegal and unethical without explicit written authorization. A dedicated, isolated lab environment solves this problem by providing a legally safe space in which an analyst can experiment freely.

A well-designed lab replicates the core components of a real enterprise network at a small scale: an attacker machine, one or more vulnerable targets, and a segmented network connecting them. This mirrors the environments that SOC analysts, penetration testers, and incident responders encounter in their daily work, allowing interns to build muscle memory for real-world scenarios in a risk-free setting.

2.2 Benefits of a Virtual Lab

- **Isolation and Safety:** Virtual machines run inside a hypervisor and can be fully isolated from the host operating system and the internet, eliminating the risk of accidental damage or legal exposure.
- **Reproducibility:** Snapshots allow the intern to save a known-good state and roll back instantly after a test breaks a system, enabling rapid experimentation.
- **Cost Efficiency:** Multiple operating systems and target machines can be run on a single physical computer, removing the need for dedicated hardware for every scenario.
- **Realistic Practice:** Deliberately vulnerable applications and machines (e.g., Metasploitable2, DVWA) are modeled on real-world vulnerabilities, giving practice that closely mirrors genuine findings.
- **Portability:** An entire lab, including all virtual machines and network configuration, can be exported, archived, or transferred to another host or intern.
- **Scalability:** Additional VMs, network segments, or vulnerable applications can be added incrementally as skills develop.

2.3 Learning Outcomes

- Understanding of hypervisor-based virtualization and its role in cybersecurity training.
- Practical experience installing and configuring Kali Linux, Windows, and deliberately vulnerable Linux distributions.
- Comprehension of virtual networking modes (NAT, Bridged, Host-only, Internal) and their security implications.
- Ability to verify network connectivity and machine reachability using command-line diagnostic tools.
- Familiarity with core security tools including Nmap, Wireshark, and Burp Suite.
- Development of structured technical documentation and reporting skills expected of a professional security analyst.

2.4 Practical Experience Gained

By the end of this task, the intern will have hands-on experience covering the full lifecycle of lab creation — from downloading and installing hypervisor software, to provisioning virtual machines, to configuring isolated networks, to verifying operational status through command-line tools. This experience directly parallels the initial setup steps performed by professional penetration testers and SOC engineers when preparing engagement environments, internal training ranges, or malware analysis sandboxes.

3. Why It Matters

Cybersecurity lab environments are not merely a training convenience — they are a core operational tool used throughout the security industry. The following table summarizes how different security roles rely on lab environments in their day-to-day work.

Role	How Lab Environments Are Used
SOC Analyst	Practices identifying malicious traffic patterns, tuning detection rules, and replaying attack traffic captures (PCAPs) in an isolated environment before applying lessons to production monitoring tools.
Penetration Tester	Rehearses exploitation techniques, tests new tools and payloads, and validates proof-of-concept exploits safely before using them against an authorized client engagement.
Incident Responder	Recreates and studies attacker techniques observed in real incidents, tests containment and eradication procedures, and trains on forensic artifact collection without risking live systems.
Security Engineer	Validates new security controls (firewalls, EDR, SIEM rules, network segmentation) against simulated attacks before deploying changes into production environments.
Malware Analyst	Detonates and observes suspicious files or binaries inside a fully isolated, network-restricted virtual machine to study behavior without risking the analyst's host or corporate network.

Across all of these roles, the common thread is risk-free experimentation. A lab environment allows security professionals to fail safely, iterate quickly, and build deep technical intuition — qualities that are essential in an industry where mistakes made in production can be costly, disruptive, or legally significant.

4. Lab Requirements

The table below outlines the hardware and software prerequisites for building the lab, along with the minimum and recommended specifications.

Hardware	Software	Purpose	Minimum Requirement	Recommended Requirement
CPU	VT-x / AMD-V enabled	Enables hardware-assisted virtualization for running multiple VMs	4-core, 64-bit, virtualization support enabled in BIOS/UEFI	8-core or higher, virtualization support enabled
RAM	Host OS + Hypervisor	Allows the host and several VMs to run concurrently without performance degradation	8 GB	16 GB or higher
Storage	SSD/HDD	Stores hypervisor, VM disk images, ISO files, and snapshots	100 GB free space (HDD)	250 GB free space (SSD)
Network Adapter	Virtual NIC (Host-only/Internal)	Provides isolated network connectivity between VMs	1 physical NIC (host)	1 physical NIC + virtual adapters configured
Display	1080p Monitor	Enables comfortable multi-window VM management	1366x768	1920x1080 or higher

In addition to hardware, the following host operating system and account privileges are required:

- A 64-bit host operating system (Windows 10/11, Linux, or macOS) with administrator/root privileges.
- A stable internet connection for downloading hypervisor installers and ISO images (required only during setup).
- Sufficient permissions to enable virtualization extensions (VT-x/AMD-V) in BIOS/UEFI if not already enabled.

5. Software Used

5.1 VirtualBox / VMware

VirtualBox (Oracle) and VMware Workstation are Type-2 hypervisors that allow multiple guest operating systems to run simultaneously on a single host machine. For this lab, VirtualBox was selected due to its free licensing, cross-platform support, and strong community documentation, though VMware Workstation Player/Pro is a suitable commercial alternative with similar functionality.

5.2 Kali Linux

Kali Linux is a Debian-based distribution maintained by Offensive Security, purpose-built for penetration testing and security auditing. It ships with hundreds of pre-installed tools spanning reconnaissance, vulnerability analysis, exploitation, and reporting, making it the standard attacker workstation for security labs.

5.3 Windows 10/11

A Windows virtual machine simulates a realistic corporate endpoint, allowing the intern to practice endpoint-focused techniques such as Windows enumeration, Active Directory basics, EDR interaction, and Windows-specific network diagnostics.

5.4 Metasploitable2

Metasploitable2 is an intentionally vulnerable Linux virtual machine distributed by Rapid7, pre-configured with numerous outdated and misconfigured services. It is widely used for practicing vulnerability scanning and exploitation in a legal, contained environment.

5.5 OWASP Broken Web Apps (BWA)

The OWASP Broken Web Applications Project is a virtual machine containing multiple deliberately insecure web applications, allowing hands-on practice with common web vulnerabilities such as SQL injection, cross-site scripting, and broken authentication.

5.6 DVWA (Damn Vulnerable Web Application)

DVWA is a PHP/MySQL-based web application containing a range of vulnerabilities at configurable difficulty levels, making it ideal for structured, progressive practice of web application security testing techniques.

5.7 Burp Suite

Burp Suite is an integrated platform for web application security testing, providing an intercepting proxy, automated scanner, and a suite of manual testing tools used to identify and exploit vulnerabilities in web applications.

5.8 Wireshark

Wireshark is an open-source network protocol analyzer used to capture and inspect network traffic in real time, essential for understanding protocol behavior and diagnosing lab connectivity.

5.9 Nmap

Nmap (Network Mapper) is a free and open-source utility for network discovery, port scanning, service/version detection, and OS fingerprinting, and is one of the most widely used reconnaissance tools in the industry.

6. Installation Process

6.1 Downloading Required Software

The following official sources should be used to download each component, ensuring file integrity through checksum verification where available:

- VirtualBox installer and Extension Pack — from the official Oracle VirtualBox downloads page.
- Kali Linux ISO or pre-built VirtualBox OVA image — from the official Kali Linux downloads page.
- Windows 10/11 ISO — from the official Microsoft Software Download portal (evaluation/developer edition is suitable for lab use).
- Metasploitable2 — from the official Rapid7/SourceForge repository.
- OWASP BWA VM — from the official OWASP project page.
- DVWA — cloned from its official GitHub repository, or deployed via the DVWA Docker image.

6.2 Installing VirtualBox

Steps to install the hypervisor on the host machine:

- Run the downloaded VirtualBox installer with administrator privileges.
- Accept the default installation path, or specify a custom directory with sufficient free space.
- Allow installation of the network adapter drivers when prompted (required for virtual networking).
- Install the VirtualBox Extension Pack separately to enable USB 2.0/3.0 support and other advanced features.
- Restart the host machine if prompted after installation.

6.3 Installing Kali Linux

Steps to provision the Kali Linux attacker VM:

- Open VirtualBox Manager and select 'New' to create a virtual machine.
- Assign a name (e.g., 'Kali-Attacker'), set type to Linux and version to Debian (64-bit).
- Allocate a minimum of 4 GB RAM and 40 GB of dynamically allocated disk space.
- Attach the downloaded Kali ISO to the virtual optical drive, or import the pre-built OVA directly via File > Import Appliance.
- Boot the VM and follow the guided installer (or first-boot wizard if using the OVA), setting a strong root/user password.
- Install VirtualBox Guest Additions inside the VM for improved display resolution and shared clipboard support.

6.4 Installing the Windows Virtual Machine

- Create a new VM in VirtualBox, selecting type Microsoft Windows and the appropriate version (10 or 11, 64-bit).
- Allocate at least 4 GB RAM and 60 GB of disk space.
- Attach the Windows ISO and proceed through the standard Windows setup wizard.
- Skip or defer a Microsoft account sign-in for an isolated lab machine where possible; create a local account instead.
- Install VirtualBox Guest Additions to enable improved resolution, drag-and-drop, and shared folders.

6.5 Installing Metasploitable2, OWASP BWA, and DVWA

- For Metasploitable2 and OWASP BWA: extract the downloaded archive and use File > Import Appliance (or New VM pointing to the included VMDK) in VirtualBox; no OS installation is required as these are pre-built images.
- For DVWA: deploy on the Kali Linux VM or a dedicated Linux VM using either a manual LAMP-stack installation or the official Docker image.

```
# Example: deploying DVWA via Docker on Kali Linux
sudo apt update
sudo apt install -y docker.io
sudo systemctl enable --now docker
sudo docker pull vulnerables/web-dvwa
sudo docker run -d -p 80:80 --name dvwa vulnerables/web-dvwa
```

After deployment, DVWA can be accessed from a browser inside the lab network at the target machine's assigned IP address, and its difficulty level configured from the DVWA Security settings page after initial login.

6.6 Configuration and Snapshots

- Configure shared clipboard and drag-and-drop settings for each VM as needed for convenience (disable for VMs intended to remain strictly isolated, such as malware analysis targets).
- Once each VM is installed, updated, and verified functional, take a snapshot (Machine > Take Snapshot) labeled clearly, e.g., 'Clean-Install-Baseline', so the machine can be reverted after destructive testing.
- Repeat snapshotting after any significant configuration milestone (e.g., 'Post-Tools-Install', 'Pre-Exploit-Test').

7. Network Configuration

VirtualBox (and VMware) provide several distinct virtual networking modes, each with different implications for isolation, connectivity, and security. Selecting the correct mode for each VM is critical to keeping the lab both functional and safely contained.

Mode	Description	Advantages	Disadvantages
NAT	The VM shares the host's network connection through a translated address; the VM can reach the internet but is not directly reachable from the host or other VMs by default.	Simple, works out-of-the-box, provides internet access for downloading updates and tools.	VMs are not visible to each other by default, which limits attacker-to-target lab scenarios unless a NAT Network is configured.
Bridged	The VM appears as a separate physical device directly on the host's physical LAN, receiving its own IP address from the local router.	VM behaves like a real machine on the physical network; useful for testing against real local devices with authorization.	Exposes the VM directly to the physical network, increasing risk if the VM is a deliberately vulnerable target; not recommended for lab targets.
Host-only	Creates a private network between the host and its VMs only; VMs can reach the host and each other, but not the internet or external network.	Strong isolation, ideal for connecting attacker and target VMs safely while still allowing host access for management.	No internet access for VMs on this adapter alone, requiring a secondary NAT adapter for updates.
Internal Network	Creates a private network visible only to the VMs attached to it; the host itself is excluded.	Maximum isolation between VMs and the host; ideal for malware detonation or highly sensitive exploitation practice.	No access to host or internet at all from this adapter, which can complicate file transfer and tool updates.

7.1 Recommended Lab Network Design

For this lab, a dual-adapter approach was used on each virtual machine: Adapter 1 configured as NAT (for internet access to download updates and tools) and Adapter 2 configured as Host-only or Internal Network (for isolated inter-VM communication between the Kali attacker machine and the vulnerable targets). This design balances the convenience of internet connectivity with the safety of a contained attack surface, ensuring that scanning and exploitation traffic never leaves the virtual lab boundary.

- Kali Linux (Attacker): Adapter 1 = NAT, Adapter 2 = Host-only (vboxnet0)
- Metasploitable2 (Target): Adapter 1 = Host-only (vboxnet0) only — no NAT, to prevent accidental internet exposure of a deliberately vulnerable machine
- OWASP BWA (Target): Adapter 1 = Host-only (vboxnet0) only
- Windows 10/11 (Endpoint): Adapter 1 = NAT, Adapter 2 = Host-only (vboxnet0)

8. Lab Verification

After configuring each virtual machine and its network adapters, connectivity must be verified before proceeding to any security testing exercises. The following commands were used to confirm correct IP addressing, routing, and reachability across the lab.

8.1 Verifying IP Configuration

```
# Windows
ipconfig /all

# Linux / Kali
ifconfig & ip a
```

These commands confirm that each VM has received a valid IP address on both the NAT and Host-only adapters, matching the expected subnet configured in VirtualBox's network settings.

8.2 Verifying Connectivity Between VMs

```
# From Kali, test reachability of the Metasploitable2 target
ping -c 4 192.168.56.102

# From Windows, test reachability of the host-only gateway
ping 192.168.56.1
```

8.3 Verifying Active Connections and Listening Services

```
# Linux / Kali
netstat -tulnp

# Windows
netstat -ano
```

8.4 Verifying ARP Table and Neighboring Hosts

```
arp -a
```

8.5 Confirming Hostnames

```
hostname
```

Successful verification is indicated by: each VM displaying an IP address in the expected Host-only subnet (e.g., 192.168.56.0/24); ping requests between the Kali attacker VM and each target VM completing with 0% packet loss; and target services (e.g., DVWA on port 80, Metasploitable2 services on various ports) appearing in netstat output or being reachable via an Nmap scan.

9. Findings

The tables below summarize the key observations recorded during the lab build and verification process.

9.1 Network Connectivity Observations

Source VM	Destination VM	Adapter Used	Result	Observation
Kali Linux	Metasploitable2	Host-only (vboxnet0)	Success	4/4 ICMP replies received; average round-trip time under 1 ms, confirming isolated segment functions correctly.
Kali Linux	OWASP BWA	Host-only (vboxnet0)	Success	Web application reachable via browser on target IP; DVWA/BWA login page rendered correctly.
Windows VM	Kali Linux	Host-only (vboxnet0)	Success	Ping and basic port checks confirmed connectivity across the private segment.
Kali Linux / Windows	Internet (via NAT)	NAT adapter	Success	Package manager and Windows Update both reached external repositories, confirming NAT adapter provides outbound internet access.
Metasploitable2	Internet	None configured	Blocked (expected)	No route to the internet exists, confirming the deliberately vulnerable target remains fully isolated from external exposure.

9.2 Nmap Scan Summary (Example Findings)

Target	Open Ports Identified	Notable Service	Risk Note
Metasploitable2	21, 22, 23, 80, 139, 445, 3306, 5432, 8180	vsftpd 2.3.4, Samba, MySQL, Tomcat	Numerous outdated services present by design; suitable for controlled vulnerability practice only.
OWASP BWA	22, 80, 443	Apache with multiple vulnerable web apps	Web root hosts several intentionally insecure applications; not to be exposed beyond the lab segment.
Windows VM	135, 139, 445, 3389	SMB, RDP	Standard Windows services; useful for practicing enumeration and RDP hardening review.

9.3 General Observations

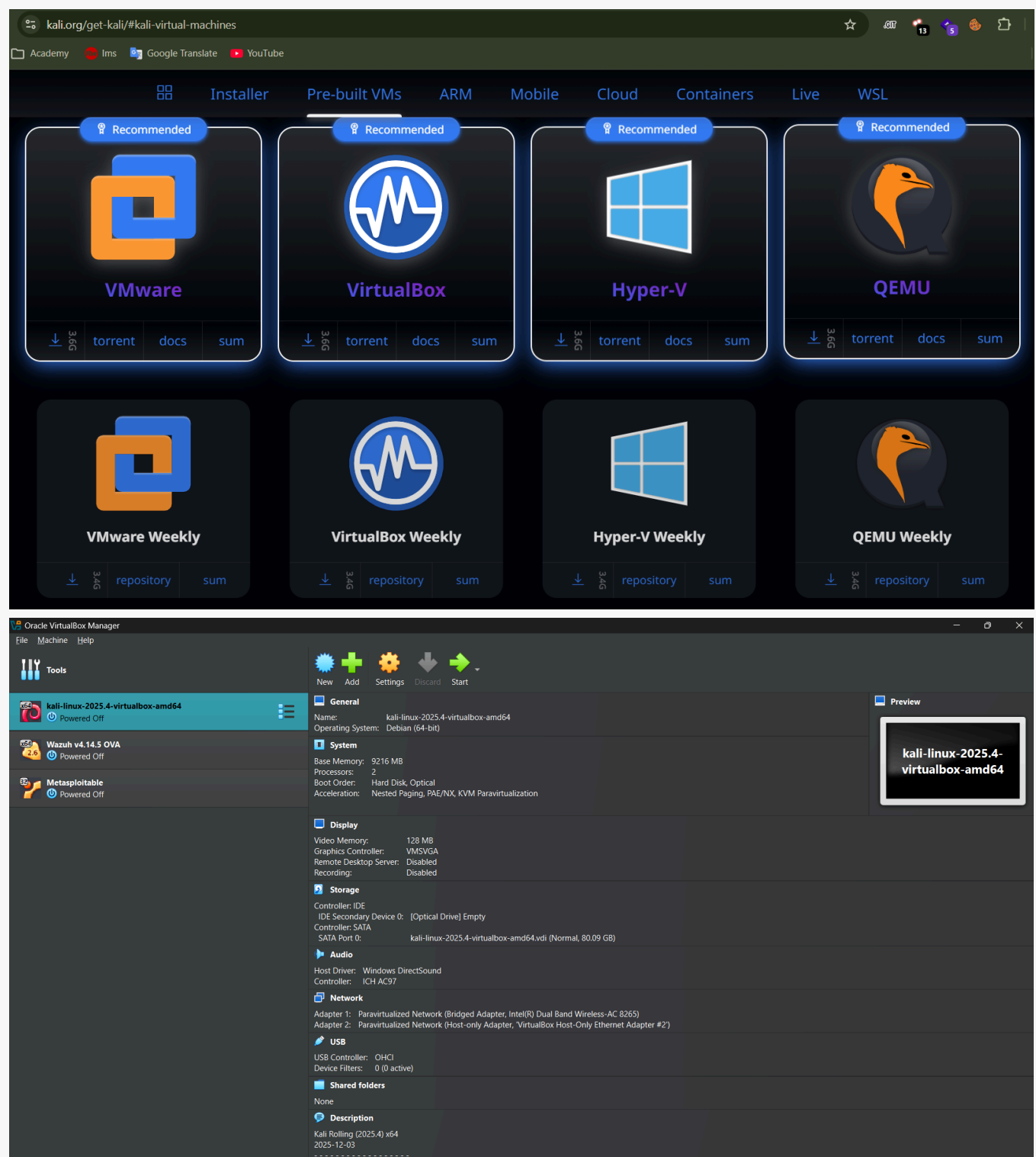
- The dual-adapter (NAT + Host-only) configuration successfully balanced internet access for updates with strict isolation for vulnerable targets.
- Guest Additions installation significantly improved usability (display scaling, clipboard sharing) across all VMs.
- Snapshotting before any exploitation activity proved essential for quickly restoring a clean environment after a target VM was intentionally compromised.
- Allocating at least 4 GB RAM per VM was necessary to avoid noticeable performance degradation when running multiple machines concurrently.

10. Screenshot Placeholders

Figure 1: Installing VirtualBox

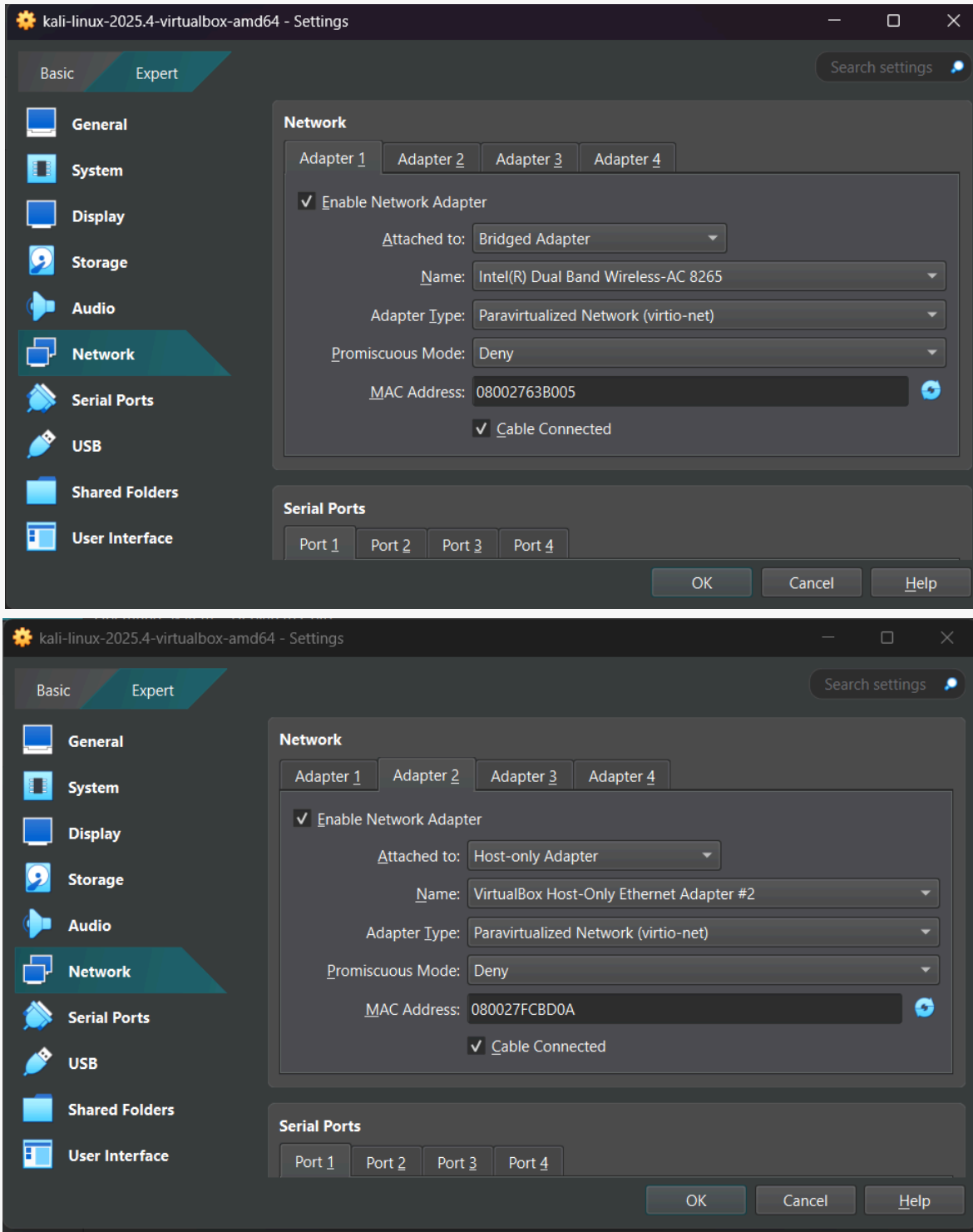


Screenshot of the VirtualBox installation wizard confirming successful installation on the host machine.

Figure 2: Installing Kali Linux

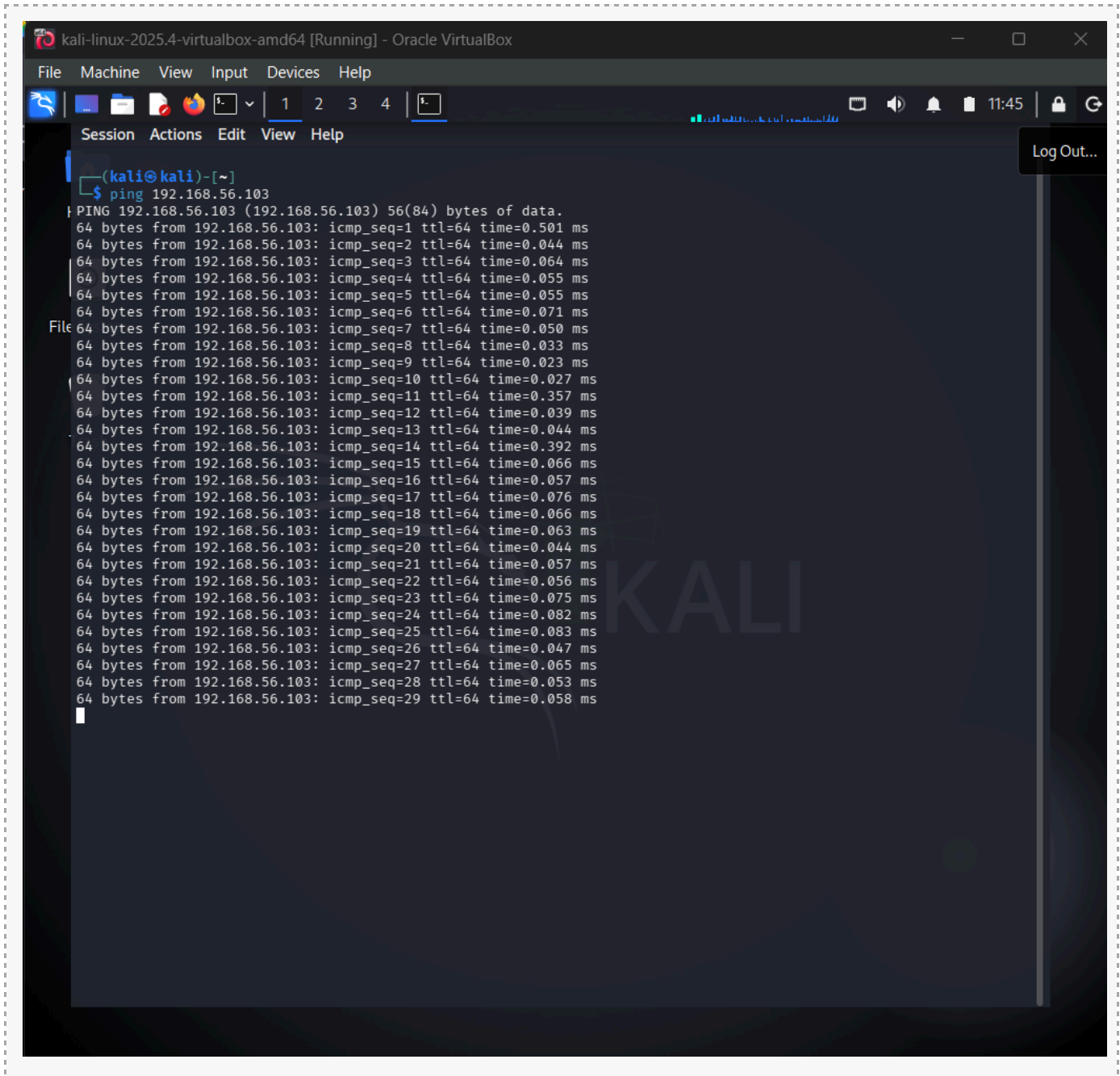
Screenshot of the Kali Linux installer/first-boot screen during virtual machine provisioning.

Figure 3: Network Configuration



Screenshot of the VirtualBox network settings panel showing NAT and Host-only adapter configuration.

Figure 4: Ping Test

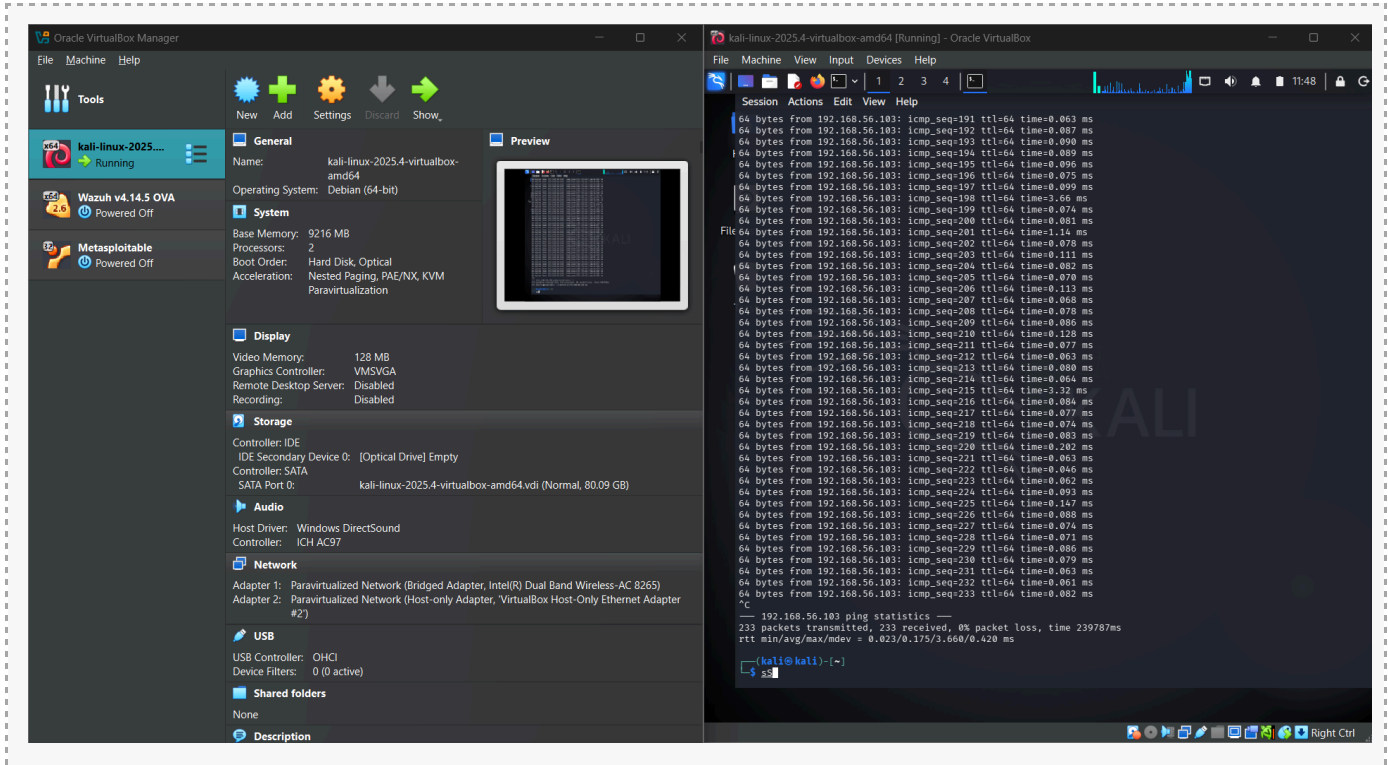


The screenshot shows a terminal window titled "kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal displays the command `ping 192.168.56.103` and its output. The output shows 30 successful ping requests, each receiving 64 bytes of data with a TTL of 64 and a response time between 0.023 ms and 0.357 ms. A large "KALI" watermark is visible in the background of the terminal.

```
(kali@kali)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.501 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.044 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.064 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.055 ms
64 bytes from 192.168.56.103: icmp_seq=5 ttl=64 time=0.055 ms
64 bytes from 192.168.56.103: icmp_seq=6 ttl=64 time=0.071 ms
64 bytes from 192.168.56.103: icmp_seq=7 ttl=64 time=0.050 ms
64 bytes from 192.168.56.103: icmp_seq=8 ttl=64 time=0.033 ms
64 bytes from 192.168.56.103: icmp_seq=9 ttl=64 time=0.023 ms
64 bytes from 192.168.56.103: icmp_seq=10 ttl=64 time=0.027 ms
64 bytes from 192.168.56.103: icmp_seq=11 ttl=64 time=0.357 ms
64 bytes from 192.168.56.103: icmp_seq=12 ttl=64 time=0.039 ms
64 bytes from 192.168.56.103: icmp_seq=13 ttl=64 time=0.044 ms
64 bytes from 192.168.56.103: icmp_seq=14 ttl=64 time=0.392 ms
64 bytes from 192.168.56.103: icmp_seq=15 ttl=64 time=0.066 ms
64 bytes from 192.168.56.103: icmp_seq=16 ttl=64 time=0.057 ms
64 bytes from 192.168.56.103: icmp_seq=17 ttl=64 time=0.076 ms
64 bytes from 192.168.56.103: icmp_seq=18 ttl=64 time=0.066 ms
64 bytes from 192.168.56.103: icmp_seq=19 ttl=64 time=0.063 ms
64 bytes from 192.168.56.103: icmp_seq=20 ttl=64 time=0.044 ms
64 bytes from 192.168.56.103: icmp_seq=21 ttl=64 time=0.057 ms
64 bytes from 192.168.56.103: icmp_seq=22 ttl=64 time=0.056 ms
64 bytes from 192.168.56.103: icmp_seq=23 ttl=64 time=0.075 ms
64 bytes from 192.168.56.103: icmp_seq=24 ttl=64 time=0.082 ms
64 bytes from 192.168.56.103: icmp_seq=25 ttl=64 time=0.083 ms
64 bytes from 192.168.56.103: icmp_seq=26 ttl=64 time=0.047 ms
64 bytes from 192.168.56.103: icmp_seq=27 ttl=64 time=0.065 ms
64 bytes from 192.168.56.103: icmp_seq=28 ttl=64 time=0.053 ms
64 bytes from 192.168.56.103: icmp_seq=29 ttl=64 time=0.058 ms
```

Screenshot of a successful ping test between the Kali Linux attacker VM and the Metasploitable2 target VM.

Figure 5: Successful Lab Setup



Screenshot of all virtual machines running simultaneously within VirtualBox Manager, confirming a fully operational lab.

11. Recommendations

The following professional recommendations should be followed to maintain a secure, reliable, and effective cybersecurity lab over time:

- Never bridge deliberately vulnerable machines (Metasploitable2, OWASP BWA, DVWA) onto a physical or shared network; restrict them to Host-only or Internal Network adapters only.
- Maintain regular snapshots before and after each testing session to allow rapid rollback in the event a target VM is corrupted or rendered unusable.
- Keep the hypervisor (VirtualBox/VMware) and attacker VM (Kali Linux) updated regularly, while intentionally leaving vulnerable target machines unpatched, since their outdated state is required for practice.
- Disable shared folders, clipboard sharing, and drag-and-drop on any VM used for malware analysis or handling untrusted files, to prevent accidental host contamination.
- Document the exact IP addressing scheme, adapter assignments, and credentials for each VM in a lab inventory sheet for repeatability and knowledge transfer.
- Periodically export lab VMs as OVA archives to enable backup, portability, and rapid redeployment on a new host.
- Enforce strong, unique passwords on the Windows VM and any VM with services exposed even within the isolated network segment.
- Review and rotate any credentials or API keys used within the lab, and never reuse lab credentials for real accounts or production systems.
- Ensure the host machine's own security software (antivirus/EDR) is configured to exclude or carefully monitor the VM disk image directories, as many contain intentionally malicious or vulnerable content that may trigger false-positive alerts.

12. Deliverables Checklist

Deliverable	Status
VirtualBox installed and verified on host machine	Completed
Kali Linux VM installed, updated, and Guest Additions configured	Completed
Windows 10/11 VM installed and configured	Completed
Metasploitable2 VM deployed and network-isolated	Completed
OWASP BWA VM deployed and network-isolated	Completed
DVWA deployed and accessible within lab network	Completed
Host-only and NAT network adapters configured on all VMs	Completed
Connectivity verified via ping, ifconfig/ipconfig, netstat, arp	Completed
Baseline snapshots captured for all VMs	Completed
Screenshots captured for all required figures	Pending attachment
Final report reviewed and submitted to NovaShyld Technologies	Completed

13. Skills Gained

Skill Category	Specific Skill Developed
Virtualization	Installing and configuring a Type-2 hypervisor; provisioning and managing multiple virtual machines.
Operating Systems	Installing and configuring Kali Linux and Windows 10/11 in a virtualized environment.
Networking	Understanding and configuring NAT, Bridged, Host-only, and Internal Network modes; subnetting within a lab environment.
Command-Line Diagnostics	Using ipconfig, ifconfig, ip a, ping, netstat, arp, and hostname to verify system and network status.
Security Tooling	Familiarity with Nmap, Wireshark, and Burp Suite as installed and available for use within the lab.
Vulnerable Application Deployment	Deploying and configuring Metasploitable2, OWASP BWA, and DVWA for safe, legal security practice.
Documentation	Producing structured, professional technical documentation suitable for internal training and audit records.
Operational Discipline	Applying snapshotting, network isolation, and credential hygiene practices consistent with real-world lab management standards.

14. GitHub Repository Structure

To support documentation, repeatability, and future reference, the lab setup artifacts (configuration notes, scripts, and this report) are organized in a version-controlled repository using the following structure:

```
cybersecurity-lab-setup/  
├── README.md  
├── docs/  
│   ├── NovaShyld_Cybersecurity_Lab_Setup_Report.docx  
│   └── network-diagram.png  
├── screenshots/  
│   ├── 01-virtualbox-install.png  
│   ├── 02-kali-install.png  
│   ├── 03-windows-install.png  
│   ├── 04-network-config.png  
│   ├── 05-ping-test.png  
│   └── 06-lab-setup-complete.png  
├── scripts/  
│   ├── dvwa-docker-setup.sh  
│   └── verify-lab-connectivity.sh  
└── config/  
    ├── vboxnet-hostonly.conf  
    └── vm-network-mapping.md
```

This structure separates final documentation (docs/), visual evidence (screenshots/), automation helpers (scripts/), and network/VM configuration references (config/), making the lab build easy to audit, reproduce, or hand off to another intern.

15. References

The following official documentation sources were used to compile this report:

- Oracle VirtualBox — Official User Manual and Downloads, [virtualbox.org](https://www.virtualbox.org)
- Broadcom / VMware — VMware Workstation Pro/Player Documentation, [vmware.com](https://www.vmware.com)
- Kali Linux — Official Documentation and Downloads, kali.org
- Microsoft — Windows 10/11 Official Documentation and Software Download Center, microsoft.com
- OWASP Foundation — OWASP Broken Web Applications Project and DVWA Reference, owasp.org
- Nmap — Official Reference Guide and Documentation, nmap.org
- Wireshark — Official User's Guide and Documentation, wireshark.org